

Overview of Universal approximation theorem

Subject: Universal approximation theorem

$$f(x) = \sum_{i=1}^N v_i \sigma(w_i^T x + b_i)$$

1.

Bounded depth and bounded width case The first result on approximation capabilities of neural networks with bounded number of layers, each containing a limited number of artificial neurons was obtained by Maierov and Pinkus. Their remarkable result revealed that such networks can be universal approximators and for achieving this property two hidden layers are enough.

2.

This is an existence result. It says that activation functions providing universal approximation property for bounded depth bounded width networks exist. Using certain algorithmic and computer programming techniques, Guliyev and Ismailov efficiently constructed such activation functions depending on a numerical parameter. The developed algorithm allows one to compute the activation functions at any point of the real axis instantly. For the algorithm and the corresponding computer code see. The theoretical result can be formulated as follows.

3.

A spate of papers in the 1980s—1990s, from George Cybenko and Kurt Hornik etc, established several universal approximation theorems for arbitrary width and bounded depth. See for reviews. The following is the most often quoted: Also, certain non-continuous activation functions such as the step function can be approximated by continuous activation functions, which then allows the approximation result to apply to those functions. In particular, this shows that a perceptron network with a single infinitely wide hidden layer can approximate arbitrary functions. Such an f $\{\displaystyle f\}$ can also be approximated by a network of greater depth by using the same construction for the first layer and approximating the identity function with later layers.

4.

Reservoir computing and quantum reservoir computing In reservoir computing a sparse recurrent neural network with fixed weights equipped of fading memory and echo state property is followed by a trainable output layer. Its universality has been demonstrated separately for what concerns networks of rate neurons and spiking neurons, respectively. In 2024, the framework has been generalized and extended to quantum reservoirs where the reservoir is based on qubits defined over Hilbert spaces.

5.

Arbitrary depth The arbitrary depth case was also studied by a number of authors such as Gustaf Gripenberg in 2003, Dmitry Yarotsky, Zhou Lu et al in 2017, Boris Hanin and Mark Sellke in 2018 who focused on neural networks with ReLU activation function. In 2020, Patrick Kidger and Terry

Lyons extended those results to neural networks with general activation functions such, e.g. tanh or GeLU. One special case of arbitrary depth is that each composition component comes from a finite set of mappings. In 2024, Cai constructed a finite set of mappings, named a vocabulary, such that any continuous function can be approximated by compositing a sequence from the vocabulary. This is similar to the concept of compositionality in linguistics, which is the idea that a finite vocabulary of basic elements can be combined via grammar to express an infinite range of meanings.

6.

In the field of machine learning, the universal approximation theorems (UATs) state that neural networks with a certain structure can, in principle, approximate any continuous function to any desired degree of accuracy. These theorems provide a mathematical justification for using neural networks, assuring researchers that a sufficiently large or deep network can model the complex, non-linear relationships often found in real-world data. The best-known version of the theorem applies to feedforward networks with a single hidden layer. It states that if the layer's activation function is non-polynomial (which is true for common choices like the sigmoid function or ReLU), then the network can act as a "universal approximator." Universality is achieved by increasing the number of neurons in the hidden layer, making the network "wider." Other versions of the theorem show that universality can also be achieved by keeping the network's width fixed but increasing its number of layers, making it "deeper." These are existence theorems. They guarantee that a network with the right structure exists, but they do not provide a method for finding the network's parameters (training it), nor do they specify exactly how large the network must be for a given function. Finding a suitable network remains a practical challenge that is typically addressed with optimization algorithms like backpropagation.

7.

The above proof has not specified how one might use a ramp function to approximate arbitrary functions in $C_0(\mathbb{R}^n, \mathbb{R})$. A sketch of the proof is that one can first construct flat bump functions, intersect them to obtain spherical bump functions that approximate the Dirac delta function, then use those to approximate arbitrary functions in $C_0(\mathbb{R}^n, \mathbb{R})$. The original proofs, such as the one by Cybenko, use methods from functional analysis, including the Hahn-Banach and Riesz–Markov–Kakutani representation theorems. Cybenko first published the theorem in a technical report in 1988, then as a paper in 1989. Notice also that the neural network is only required to approximate within a compact set K . The proof does not describe how the function would be extrapolated outside of the region. The problem with polynomials may be removed by allowing the outputs of the hidden layers to be multiplied together (the "pi-sigma networks"), yielding the generalization:

8.

Variants Variants include discontinuous activation functions, noncompact domains, certifiable networks, random neural networks, and alternative network architectures and topologies. The universal approximation property of width-bounded networks has been studied as a dual of classical universal approximation results on depth-bounded networks. For input dimension d_x and output dimension d_y the minimum width required for the universal approximation of the L_p functions is exactly $\max\{d_x + 1, d_y\}$

$\max\{d_{\{x\}}+1, d_{\{y\}}\}$ (for a ReLU network). More generally this also holds if both ReLU and a threshold activation function are used. Universal function approximation on graphs (or rather on graph isomorphism classes) by popular graph convolutional neural networks (GCNs or GNNs) can be made as discriminative as the Weisfeiler–Leman graph isomorphism test. In 2020, a universal approximation theorem result was established by Brüel-Gabrielsson, showing that graph representation with certain injective properties is sufficient for universal function approximation on bounded graphs and restricted universal function approximation on unbounded graphs, with an accompanying $O(|V| \cdot |E|)$ -runtime method that performed at state of the art on a collection of benchmarks (where V and E are the sets of nodes and edges of the graph respectively). There are also a variety of results between non-Euclidean spaces and other commonly used architectures and, more generally, algorithmically generated sets of functions, such as the convolutional neural network (CNN) architecture, radial basis functions, or neural networks with specific properties.